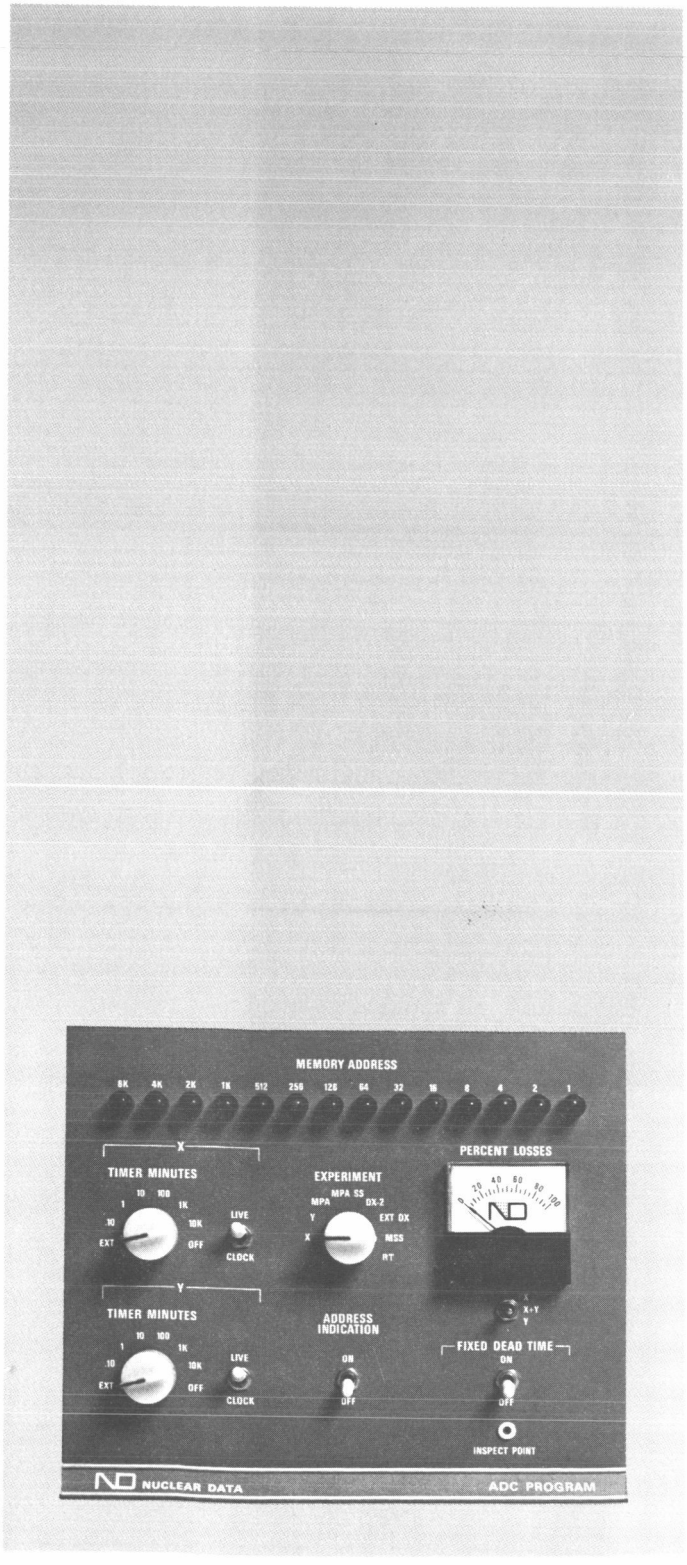
TECHNICAL SPECIFICATIONS

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SERIES 3300 ADC PROGRAM

The ADC Program Module establishes the logical conditions for a variety of operations with either one or two Analog-to-Digital Converter Modules (ADC's). Operations which can be selected include single parameter analysis with either the X ADC or Y ADC, two parameter analysis with both the X and Y ADC's, two parameter operation with periodic singles spectra sampling, time-shared single parameter analysis (digiplex) with both ADC's, externally controlled digiplex, spectrum multiscaling and routing.

Two time reference oscillators, one for the X ADC and the other for the Y ADC, are provided for experiment timing. Either clock or live timing, with a wide range of reference units, can be selected. A fixed dead time for each ADC conversion gain factor can also be selected, if desired.

Memory address lamps are included to provide a binary indication of storage location. In addition, dead time indication is provided by a percent losses meter. A front panel control enables selection of dead time indication for either the X ADC, the Y ADC, or a combination of both.

ADC Operation

X: Single parameter analysis with X ADC.

Y: Single parameter analysis with Y ADC.

MPA: Two parameter analysis with both X and Y ADC's.

Singles Sampling: Two parameter analysis with periodic sampling of singles from X ADC and Y ADC, with singles stored in X=0 plane and Y=0 plane, respectively.

Digiplex 2: Single parameter analysis, with time-shared memory storage for data from the X ADC and the Y ADC.

External Digiplex: Time-shared single parameter analysis with external control.

Spectrum Multiscaling: Time-sequenced single parameter analysis, with separate storage for the spectrum acquired during each dwell period.

Routing: Single parameter analysis with time-shared ADC and separate storage for data from the X ADC for up to four inputs.

X & Y Time Units

0.1, 1, 10, 100 1K, or 10K minutes. An external time base can also be introduced, or the internal reference oscillator can be turned off, if desired.

Time Reference Stability

24 hours; Less than 0.01% deviation at constant ambient temperature.

Temperature: Less than 0.1% deviation per 10°C increase from 5°C to 45°C.

Live Time Accuracy

Better than 0.5% at input count rates of 5,000 pulses per second, with experiment times for adequate probability statistics.

Memory Address Indication

14 bits, scale of 16,384-1.

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